

HSBC CreditLens turns AI credit scoring into jurisdiction-specific compliance evidence

Why now

AI credit models create explainability, fairness and governance questions that differ sharply between the US and Hong Kong.

Product

A jurisdiction-aware dashboard that reads model outputs, applies local rule parameters, and produces management-ready alerts.

Boundary

It does not make legal conclusions or replace credit officers; it makes the risks visible enough for accountable judgement.

A global bank needs one model-control language without pretending regulation is the same everywhere

Entity fit

HSBC has material operations in Hong Kong and the United States, making it a realistic cross-border RegTech client.

Domain fit

Consumer credit scoring is high-impact, data-rich, and exposed to fairness, explainability, privacy and governance expectations.

Management fit

Senior leaders need a concise view of where the model is performing, where it is unfair, and where local rules change the answer.

The US asks 'can you explain the denial?' while Hong Kong asks 'is the AI governed proportionately?'

US lens

ECOA / Regulation B: specific principal reasons for adverse action, fair lending screening, and defensible use of complex algorithms.

Hong Kong lens

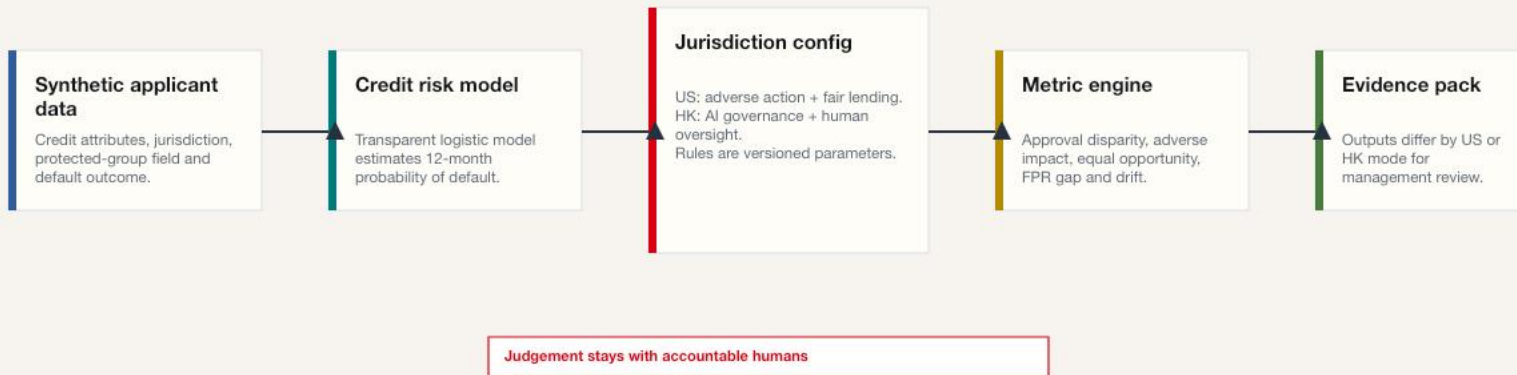
HKMA and PCPD: accountability, validation, human oversight, transparency, privacy and consumer credit data controls.

Design implication

The same score should produce different evidence packs depending on the jurisdiction switch.

A rule configuration layer sits between model outputs and compliance evidence

The model is deliberately separate from jurisdiction logic so rule changes do not require model rebuilds.



Synthetic-data prototype shows the metrics senior management would monitor

These are monitoring signals, not legal conclusions.

MODEL AUC

0.671

Transparent logistic baseline

APPROVAL RATE

26.4%

PD threshold set at 0.34

US ADVERSE IMPACT

0.57

Screen against 0.80 floor

HK ADVERSE IMPACT

0.54

Governance trigger at 0.85

What management learns

The model can look statistically useful while still requiring fairness review. The jurisdiction layer changes the action: US mode asks whether a denied applicant received specific reasons; Hong Kong mode asks whether governance controls are proportionate and evidenced.

US mode converts denied applications into adverse-action and fair-lending evidence

Adverse action

For denied applicants, the tool extracts top model drivers and maps them to specific principal denial reasons.

Fair lending screen

Approval-rate disparity and adverse impact ratio are calculated by protected group.

Escalation

Near-threshold or disparity-triggering decisions are routed for human review.

Hong Kong mode emphasizes governance, proportionality and human oversight

Governance evidence

Model validation, outcome monitoring, human review bands and privacy assessment are required artefacts.

Consumer credit data

The tool records whether data use aligns with consumer credit data and PDPO-oriented expectations.

Senior accountability

Outputs are framed for management review rather than as automated legal sign-off.

The product is most valuable when it admits where it can fail

Rule change

A mid-period rule update can invalidate prior evidence unless rule versions are logged.

Model drift

A model can pass validation and later create group disparity as applicant mix shifts.

Misconfiguration

Applying US evidence templates in Hong Kong, or vice versa, creates false comfort.

The tool improves governance because it refuses to automate judgement away

Does not do

No production credit decisioning, no legal opinion, no guarantee of compliance, no replacement for validation.

Does do

Turns model outputs into jurisdiction-specific metrics, explanations, review triggers and governance artefacts.

Next build

Connect to model inventory, add SHAP/LIME, version rule libraries, and integrate legal sign-off workflow.

Approve a controlled pilot for HSBC consumer credit model governance

Pilot scope

Use synthetic and historical validation data first; avoid live customer decisioning until controls are tested.

Success criteria

Specific US denial reasons, HK governance evidence pack, drift alerts and board-readable reporting.

Decision

Proceed with a 10-12 week proof of concept owned jointly by Credit Risk, Compliance, Model Risk and Legal.